

Programme	BSc Mathematics Honours			
Course Code	MAT1VN102			
Course Title	STATISTICS FOR DATA SCIENCE			
Type of Course	Vocational Minor – Introduction to Data Science			
Semester	I			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Foundations in mathematics			
Course Summary	Course aims to provide basic concepts such as central tendency, probability, sampling and testing			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand measures of central tendency , dispersion, regression	U	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO2	Distinguish discrete and continuous distributions and its properties	U	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO3	Analyse data using testing hypothesis	An	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Descriptive statistics	11	Min.15
	1	Measures of central tendency: - mean, median, mode		
	2	Measures of dispersion: Range, Mean deviation, Quartile deviation and Standard deviation		
	3	Moments, Skewness and Kurtosis,		
	4	Correlation - Linear correlation		
	5	Karl Pearson's coefficient of Correlation, Rank correlation		
	6	Linear regression- Simple and Multiple		
II		Probability	7	Min.15
	7	Sample space, Events, Different approaches to probability		
	8	Addition and multiplication theorems on probability		
	9	Independent events, Conditional probability		
	10	Bayes Theorem		
III		Probability Distributions	12	Min.15
	11	Random variables, Probability density functions and distribution functions		
	12	Marginal density functions, Joint density functions		
	12	Mathematical expectations		
	14	Moments and moment generating functions		
	15	Discrete probability distributions – Binomial, Poisson distribution		
	16	Continuous probability distributions- uniform distribution and normal distribution.		
III		Sampling and Testing		
	17	Theory of Sampling: - Population and sample, Types of sampling Theory of Estimation: - Introduction, point estimation		

	18	methods of point estimation-Maximum Likelihood estimation and method of moments, Central Limit Theorem(Statement only)	15	Min.15
	19	Null and alternative hypothesis, types of errors, level of significance, critical region		
	20	Large sample tests – Testing of hypothesis concerning mean of a population and equality of means of two populations		
	21	Small sample tests – t Test for single mean, difference of means. Paired t-test		
	22	Chi-square test (Concept of test statistic χ^2/σ^2), F test - test for equality of two population variances		
	23	ANOVA – one-way & two-way classification		
		Practical using MS Excel	30	
		1. Calculate the mean, median, and mode of a dataset. 2. Calculate the range of a dataset. 3. Calculate the mean deviation of a dataset. 4. Calculate the quartile deviation of a dataset. 5. Calculate the standard deviation of a dataset. 6. Calculate skewness and kurtosis of a dataset. 7. Compute the Karl Pearson's coefficient of correlation between two variables. 8. Calculate rank correlation (e.g., Spearman's rank correlation) between two variables. 9. Perform simple linear regression analysis. 10. Perform multiple linear regression analysis. 11. Calculate probabilities of events using different approaches (e.g., classical, relative frequency, subjective). 12. Apply addition and multiplication theorems of probability to solve problems. 13. Calculate conditional probabilities and use Bayes' Theorem. 14. Generate random samples from various probability distributions (e.g., binomial, Poisson, normal) and calculate relevant statistics. 15. Conduct hypothesis testing using Excel functions for large sample tests (e.g., z-test, t-test), small sample tests (e.g., t-test for single mean, paired t-test), chi-square test, F-test, and ANOVA.		
		References		
	1	Fundamentals of statistics: S. C. Gupta, 6th Revised and enlarged edition April 2004, Himalaya Publications		

	2	Fundamentals of Mathematical Statistics- S. C. Gupta,V. K. Kapoor. Sultan Chand Publications		
	3	Introduction to Mathematical Statistics - Robert V. Hogg & Allen T. Craig. Pearson education		
	3	Probability and Statistics for Engineering and the Sciences, Jay L. Devore, Cengage Learning, January 2022, ISBN for the 10th Edition: 978-1305251809		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	1	3	3	3	1	2
CO 2	2	1	3	1	3	3	3	1	2
CO 3	3	2	3	2	3	3	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓